

Static Signed Consensus on Every Finite Undirected Graph: Exact Semigroup Limits and the Full Pseudoforest Characteristic Polynomial

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Abstract

We treat arbitrary finite, disconnected, static, undirected simple signed graphs with strictly positive edge weights. Structural balance determines the kernel of the signed Laplacian component by component, yielding an explicit orthogonal semigroup limit and an exact operator-norm convergence rate. The main combinatorial step expands every principal minor by components that are either one-root trees or root-free negative unicycles; summing root choices gives the complete characteristic polynomial, with a factor four for each negative unicycle. A bridge-plus-negative-triangle example disproves a tree-only minor formula, and a nonsymmetric example marks the directed boundary. Exhaustive exact checks cover all 760 signed simple graphs and all 11,894 root sets through four vertices, but the arbitrary-size positive-weight theorem is proved by quadratic forms and Cauchy–Binet, not by enumeration.

Keywords: signed graph; structural balance; Altafini consensus; signed Laplacian; matrix-tree theorem; pseudoforest.

中文摘要

本文研究任意有限、可不连通、静态无向简单符号图，边权严格为正。结构平衡逐连通分量决定符号拉普拉斯矩阵的核，从而得到显式正交半群极限和精确算子范数收敛率。进一步，本文用恰含一个根的树与无根负单圈分量展开全部主子式，并由根集求和得到完整特征多项式；每个负单圈贡献因子四。四阶以内的穷举只作实现校验，不代替任意规模的证明。

关键词：符号图；结构平衡；符号一致性；拉普拉斯矩阵；伪森林。

1 Frozen graph and source ownership

Let $G = (V, E)$ be a finite undirected simple graph, not necessarily connected. Every edge $e = \{i, j\}$ has sign $\sigma_e \in \{\pm 1\}$ and weight $w_e > 0$. Choose any orientation and set

$$b_e = \mathbf{e}_i - \sigma_e \mathbf{e}_j, \quad B = [b_e]_{e \in E}, \quad W = \text{diag}(w_e), \quad L = BWB^\top.$$

Changing an edge orientation only negates a column of B and leaves L unchanged. The frozen dynamics is

$$\dot{x} = -Lx. \quad (1)$$

Structural balance is classical [2]; the antagonistic-consensus context is due to Altafini [1]; signed incidence/matrix-tree foundations and their erratum are recorded in [3, 4]. We claim no priority for those results.

2 Kernel, projector, and exact semigroup rate

For every $x \in \mathbb{R}^V$,

$$x^\top Lx = \sum_{e=ij} w_e (x_i - \sigma_e x_j)^2. \quad (2)$$

A connected component C is balanced exactly when there is a switch vector $s_C \in \{\pm 1\}^C$ satisfying $(s_C)_i = \sigma_{ij}(s_C)_j$ on every edge. An isolated vertex is balanced by this definition.

Theorem 1 (componentwise signed consensus). *The nullity of L equals the number of balanced connected components. Its orthogonal kernel projector is*

$$P = \sum_{C \text{ balanced}} \frac{s_C s_C^\top}{|C|}, \quad (3)$$

where vectors are extended by zero outside C . Moreover

$$e^{-tL} \longrightarrow P \quad (t \rightarrow \infty). \quad (4)$$

If L has positive spectrum and $\gamma = \min(\text{spec } L \setminus \{0\})$, then for every $t \geq 0$,

$$\|e^{-tL} - P\|_2 = e^{-\gamma t}. \quad (5)$$

If $L = 0$, $P = I$ and the left side is identically zero.

Proof. Equation (2) shows that $x \in \ker L$ exactly when every edge constraint $x_i = \sigma_{ij} x_j$ holds. On a connected balanced component, propagation from one vertex gives precisely $\text{span}\{s_C\}$. On an unbalanced component, propagation around a negative cycle gives $x_i = -x_i$, hence the only solution is zero. Disjoint supports make the normalized switch vectors orthonormal, proving (3). Orthogonal diagonalization of the real symmetric positive-semidefinite L proves (4); the largest remaining exponential multiplier is exactly $e^{-\gamma t}$, proving (5). $\square$

Thus each balanced component reaches signed consensus $s_C(s_C^\top x_0)/|C|$, while every unbalanced component tends to zero. No global connectedness or global balance label is required.

3 Every principal minor: rooted trees plus negative unicycles

For $R \subseteq V$, write $L[V \setminus R]$ for the principal submatrix on the nonroots. Let $\mathcal{F}_R$ be the spanning edge sets $F \subseteq E$ whose connected components are of exactly two permitted kinds:

1. a tree containing exactly one vertex of R ;
2. a root-free unicyclic component whose unique cycle has negative sign.

Write $u(F)$ for the number of components of the second kind and $w(F) = \prod_{e \in F} w_e$.

Theorem 2 (all principal minors). *For every root set, including $R = \emptyset$ and $R = V$,*

$$\det L[V \setminus R] = \sum_{F \in \mathcal{F}_R} 4^{u(F)} w(F). \quad (6)$$

Proof. Put $U = V \setminus R$. Cauchy–Binet applied to $L[U] = B[U]WB[U]^\top$ gives

$$\det L[U] = \sum_{S \subseteq E, |S|=|U|} \det(B[U, S])^2 \prod_{e \in S} w_e.$$

The incidence minor is nonzero precisely when every selected-edge component is a tree with one deleted row (one root), or has no deleted row and is an unbalanced unicycle. Leaf elimination makes a rooted-tree determinant ± 1 . Eliminating trees attached to a unicycle reduces it to its signed cycle incidence matrix, whose determinant is 0 for a positive cycle and ± 2 for a negative cycle. Squaring and multiplying components gives $4^{u(F)}$, proving (6). The empty 0×0 determinant and empty forest both equal one. $\square$

4 The complete characteristic polynomial

Let $\mathcal{F}$ consist of spanning pseudoforests whose components are trees or negative unicycles. Let $t(F)$ count the tree components.

Corollary 1 (full expansion).

$$\det(\lambda I + L) = \sum_{F \in \mathcal{F}} 4^{u(F)} w(F) \lambda^{t(F)} \prod_{T \in \text{Tree}(F)} |V(T)|. \quad (7)$$

Proof. The coefficient identity $\det(\lambda I + L) = \sum_{R \subseteq V} \lambda^{|R|} \det L[V \setminus R]$ sums all principal minors. For fixed $F \in \mathcal{F}$, every tree component must choose exactly one root, giving $|V(T)|$ choices, while negative unicycles choose none. Hence $|R| = t(F)$ and (7) follows. $\square$

The formula simultaneously contains the determinant (no tree components), all cofactors and every intermediate coefficient. It also makes every principal minor nonnegative, consistently with $L \succeq 0$.

5 Two boundary counterexamples

Consider the positive bridge 0–1 and the triangle 1–2–3–1 with signs $-, +, +$. Its signed Laplacian is

$$L = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & 1 & -1 \\ 0 & 1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{pmatrix}.$$

Here $\det L = 4$, but deleting root 0 gives determinant 7. Three comes from the ordinary spanning trees (the bridge and two triangle edges); four comes from the selected negative triangle plus the isolated root. Thus an all-root-set tree-only formula is false.

Symmetry is essential to Theorem 1. The directed matrix

$$A = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$$

has right zero vector $(1, 1)^\top$ but normalized left zero vector $(2/3, 1/3)$. Its limit projector has two identical rows $(2/3, 1/3)$ and is not the orthogonal average. Directed, time-varying, nonlinear, nonpositive-weight and self-loop models are outside the frozen theorem.

Revision focus. Round 1 adds all principal minors, the full characteristic pseudoforest expansion and two sharp boundary counterexamples.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the manuscript; no observational network data are used.

Ethics. No human, animal, personal or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported derivation, drafting and exact-code development; it was not an external reviewer or an independent peer-review process.

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